

BST Completeness — Rubric, Scorecard & Roadmap

Reference document. Keeper, 2026-08-15. Drive against this for the next few months. Purpose: one page that answers “what does it mean for BST to be *complete*, where are we, and what’s left.” Two rubrics: the external (skeptic-facing, ChatGPT/Tegmark bar) and the internal (BST-native). Both must be satisfied. Honesty is the argument — tier every cell.

1. The two rubrics

External rubric (the skeptic’s bar — keep verbatim for outreach)

The 5 items a hostile referee (Tegmark’s lens) applies to any “ambitious theory”: 1. State the fundamental postulates. 2. Derive QM. 3. Derive the SM parameters. 4. Recover GR. 5. Make new quantitative predictions.

Internal rubric (BST-native — “reasonable but sufficient”: 5 criteria, each already a BST artifact)

- A. FORCED OBJECT — one geometry (D_{IV}^5), five integers, zero dimensionless free parameters; is the geometry *itself* compelled? (*artifact: Strong-Uniqueness Theorem*)
- B. EVERYTHING IS A READING — QM, GR/spacetime, and the SM are faces of one operator, not three separate derivations. (*artifact: the one-page minimal generating set — one object + ~5 operations*)
- C. COMMITMENT ONTOLOGY — measurement-as-commitment (Principle #16) is the mechanism; information↔matter (bulk-boundary/Hardy) is how it works. (*artifact: the QM / Measurement-as-Commitment papers — the one criterion not yet a single named artifact*)
- D. FORCED, NOT FITTED — every result carries a tier (D/PD/I/C/S), target-innocent, anti-numerology. (*artifact: the tier system + the “Forcing + Evidence” reviewer paper*)
- E. COMPLETE & FALSIFIABLE — mathematically complete (uniqueness; Millennium attempts honest-tiered) and sharply falsifiable (the Six Absences + near-term kill-conditions). (*artifacts: Strong-Uniqueness, the Millennium ledger, the Five/Six-Absence set*)

Why two, and how they map

The external rubric is a generic “recover known physics” checklist — necessary, but it is additive (QM, SM, GR as separate items), has no slot for the ontology, and doesn’t score forced-vs-fitted. The internal rubric fixes exactly those three gaps. They map cleanly (the external is the projection of the internal onto standard physics):

External	Internal
1 postulates	A (forced object)
2 QM + 3 SM + 4 GR	B (all faces of one operator)
— (missing)	C (ontology)
— (missing)	D (forced-not-fitted)
5 predictions	E (falsifiable half) + math-completeness half

2. Where we are — dual scorecard (honest tiers, post-2026-08-15)

External rubric

#	Item	Status	Note
1	Postulates	STRONG (exceeds)	one geometry, 5 integers, 0 free params — shorter than GR. Residual: is D_{IV}^5 forced (K943).
2	Derive QM	DONE	10/10 Dirac-von Neumann axioms, zero posits (Code-Forces-Fermion closed item 10). The standout.

#	Item	Status	Note
3	Derive SM params	MORE BANKED THAN “FRONTIER” (re-tiered 2026-08-16, K1613)	DERIVED: α, $\theta_{\text{QCD}}=0$, $\sin^2\theta_W$, $m_p/m_e=6\pi^5$; the generation ADDRESSES $\nu=\{5/2,3/2,0\}$ (T2517, forced 3 ways); down+charged-lepton masses (forward closed forms T2003); Cabibbo $V_{us}=1/\sqrt{20} + m_s/m_d=20$ (blind, T2529/30). IDENTIFIED: Koide $Q=2/3$ (relation holds exact) — the reproducing-kernel-norm LADDER route is FALSIFIED (K1619, razor FAIL: tau boundary diverges $\rightarrow A^2=4$). “$A^2=\text{rank}$” RETIRED (rank-2 coincidence). Web-reconnect: Koide’s $2/3 = \sqrt{\text{mass vector at } 45^\circ}$ (= the equal-norm reformulation, confirmed by the literature), and its SOURCE is the mass-MATRIX at the midpoint of its range (Z_3-democratic) — the literature-endorsed route BST has NOT tested. Koide is a genuinely open problem in physics; redirected, not dead. OPEN: up-masses (saturation, not FK), the mixing fire (rides the forced addresses). Narrower honest frontier than the label; still highest over-claim risk. Count Koide+T190+T2003 as TWO numbers, not four (Cal §542).

#	Item	Status	Note
4	Recover GR / spacetime	ADVANCED (was framework)	induced Einstein eq, G formula, m_e 0.03%; + today: signature matter-induced (T2565), Lorentz via covariant-construction+matter-frame, continuum $\rightarrow \hbar$. Owed: Koons-tick scale, field-eq rigor (#114).
5	New predictions	STRONG	Six Absences, DESI $w_a > 0/D_H$, θ_{23} octant, Bell $1/2^N N_c$, $r \approx \alpha^2$, $\Sigma m_\nu(m_1=0)$, H_0 12/13. Owed: formalized register + falsifier tracking.

Internal rubric

Crit	Status	Note
A. Forced object	STRONG, residual REDUCED (2026-08-16, K1597)	Strong-Uniqueness banked. The “is D_{IV}^5 forced” residual is no longer a fog: commitment + isotropy forces TYPE-IV (Lorentzian spin factor, all other Jordan families excluded target-innocently via the cubic invariant); the isotropy principle = the T2565 spacetime keystone (one principle, double duty); the whole thing reduces to “is a commitment binary (two-outcome)?”; the dimension $n=5$ is separately open ($n \geq 5$; needs minimality). PARTIAL-CONDITIONAL forcing — a real advance. Never “commitment forces D_{IV}^5 .”
B. Everything a reading	QM done · GR advanced · SM frontier	the unification is real (QM=axioms, GR= a_1 of the <i>same</i> operator). Artifact (one-page #66) in progress.

Crit	Status	Note
C. Commitment ontology	ARTIFACT NOW EXISTS (2026-08-16, K1607/K1609)	The dig produced it: measurement-as-commitment is now ONE graded operator on D_{IV}^5 — $C = P_{record} \oplus P_{encode}$ (irreversible binary write $\oplus$ confined 3-encode), consistent, minimally sufficient (for an integrity-checked irreversible record), complete for QM. rank=2 forced externally (first ever). Owed: write it up as the #67/#66 paper; scope “complete” (operation vs geometry); $N_c=3$ map + $n_C=5$ stay open/flagged. Never “unforgeable” (self-validating).
D. Forced, not fitted	STRONG	the D/PD/I/C/S tier system is mature and operational; target-innocence discipline standing. Consolidating artifact = the “Forcing + Evidence” reviewer paper (#31).
E. Complete & falsifiable	Falsifiable STRONG · math-complete PARTIAL	Six Absences sharp; Strong-Uniqueness banked; Millennium = honest <i>attempts</i> , not consensus proofs (flagged, not a defect).

The shape is the credibility signal: QM done (rare), postulates/predictions/forcing strong, GR advanced, SM the honest frontier. A crank claims all cells 100%; BST tiers them precisely.

3. The task list to complete BOTH rubrics (next few months)

Ordered by weight. Each item tagged with the rubric cell(s) it closes and the existing board task(s).

TIER 1 — the big blocks (do these to move the needle)

1. SM-parameter frontier — *closes External 3 / Internal B*. The largest lift.
 - O7: gen-2,3 K-type addresses (#69) → unlocks the mixing sector + θ_{23} octant.
 - The fermion-mass tower protocol (K1205) → masses forced-or-falsify.
 - Charged-lepton module (#37) · Koide gate (#64) · up-quark m_u .
 - The mixing sector (7 params) via the FK overlap / directed product.
2. Finish spacetime/GR + publish — *closes External 4 / Internal B*.
 - Write up + publish the spacetime pair (T2565 + reality obstruction + constructive narrative) — Lyra PDF, Keeper ledger audit (K1571).
 - Pin the Koons-tick scale (the one owed number).
 - Crown jewel: spectral action = causal action (#114) — the GR-rigor summit.

TIER 2 — consolidate the criteria the external rubric misses

3. Consolidate the ontology (Internal C) — write the QM / Measurement-as-Commitment / Architecture papers (#67); the lifecycle-narrative capstone (#71). (*C has no artifact yet — this creates it.*)

4. Consolidate forced-not-fitted (Internal D) — the “Forcing + Evidence” reviewer paper (#31, #25). (*D as a standalone artifact = the Tegmark spine.*)
5. The forced-object residual (External 1 / Internal A) — does commitment/geometry get forced (K943/K1150, task #79)? The deepest “why this geometry.”

TIER 3 — completeness & the front door

6. Predictions register (External 5 / Internal E-falsifiable) — formalize with tiers + kill-conditions (Grace); falsifier scrub v1.0 (#107).
7. Math-completeness (Internal E-math) — the Millennium per-problem honest-tiered ledger; finalize Strong-Uniqueness.
8. The package (presents the whole) — the one-page minimal generating set (#66) + the Tegmark front-door: *“QM fully derived from one geometry (verifiable on GitHub), postulates minimal, predictions sharply falsifiable, GR a credible induced framework, SM values honestly-tiered.”*

Near-term “done” definition

BOTH rubrics are complete when: External 3 is no longer “frontier” (the mixing sector + mass tower resolve or falsify), External 4 is published with the tick-scale pinned, Internal C and D each have their named artifact (#67, #31), the forced-object residual (A) is settled, and the one-page/Tegmark package (8) presents it. Items 2 (QM) is done; 4 took a big step 2026-08-15.

4. Standing notes

- Honesty is the argument. Every external-facing cell carries its tier; never claim a “frontier” cell as done. (Item 3 is the standing over-claim risk.)
- Nothing external without Keeper-PASS + Cal-vet + both voices + Casey GO. CP existence-only. Never lead with $\alpha=137$.
- This doc supersedes the scattered scorecards (K1162, K1206) as the single reference. Update the status columns as cells close.

— Keeper, 2026-08-15. Rubric + dual scorecard + roadmap. Reference for the next few months.